

CHE 435/525 *Process Systems Analysis and Control*

Lecture 9: **Higher-Order Systems, Poles, Delays**

Wentao Tang

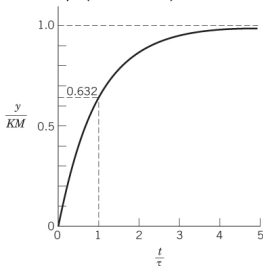
Assistant Professor, Chemical & Biomolecular Engineering
North Carolina State University

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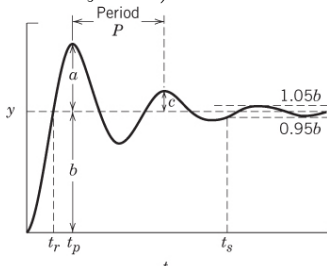
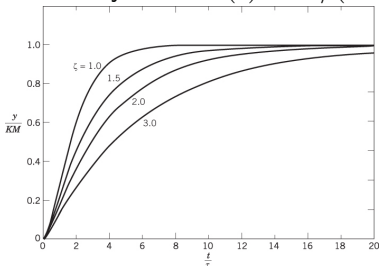


Recapture: First-order systems and second-order systems

- First-order system: $G(s) = K/(\tau s + 1)$



- Second-order system: $G(s) = K/(\tau^2 s^2 + 2\tau\zeta s + 1)$



Learning objectives of this lecture

- ① Use Matlab to **simulate** the step response of higher-order systems
- ② Precisely describe the relation between **poles** and **stability** of T.F.s
- ③ Explain how **time delay** arises in process dynamics, and describe response of delayed systems

Higher-order system

- General form of an n -th order system

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

where $a_n \neq 0$

- For any **physically realizable system**, degree of denominator $n = \deg A(s) \geq m = \deg B(s)$ degree of numerator
- **Imagine: What kind of processes can be n -th order?**
 - ▶ A system that needs to be described by n simultaneous diff. eqs. (e.g., many tanks connected, multiple reactions, a distillation column with many trays, etc.)
 - ▶ **Rule-of-thumb: Processes that involve reactions or phase equilibria are usually high-order**

Step response of higher-order systems

- Can be hard to calculate by hand ($u(t) \rightarrow U(s) \rightarrow Y(s) \rightarrow y(t)$)
- Matlab: creating a transfer function $G(s) = \frac{3s+2}{s^2+4s+5}$

```
sys = tf([3, 2], [1, 4, 5]);
```

- Matlab: simulating **unit step** response

```
[y, tOut] = step(sys);
```

- ▶ Only “step(sys)” will generate a plot without the data

- Matlab: simulating **unit impulse** response

```
[y, tOut] = impulse(sys);
```

- Matlab: simulating response to **any given input** (say, $u(t) = 8 \sin(\pi t)$)

```
t = 0:0.01:10; u = 8*sin(pi*t);  
[y, tOut] = lsim(sys, u, t);
```

"PKZ" analysis of general TFs

- "P" = poles, "K" = static gain, "Z" = zeros

Terminology

For a TF model $G(s) = \frac{B(s)}{A(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$, **poles** refer to the roots of $A(s)$ (denominator), **zeros** refer to the roots of $B(s)$ (numerator).

- If p makes $A(p) = 0$, then $|G(p)| \rightarrow \infty \dots$ hence called a "pole".
- If z makes $B(z) = 0$, then $|G(z)| \rightarrow 0 \dots$ hence called a "zero".
- Example 1: For $G(s) = \frac{\tau_0 s + 1}{(\tau_1 s + 1)(\tau_2 s + 1)}$, what are poles and zeros?
- Example 2: For $G(s) = \frac{s^2 - 3s + 2}{s^2 + 2s + 2}$, what are poles and zeros?

Effect of **poles** on **stability**

- Consider unit step input $U(s) = 1/s \Rightarrow Y(s) = B(s)/sA(s)$
 - ▶ If **all** roots of $A(s)$ have negative real parts, then $Y(s)$ ultimately settles. (All remaining terms exponentially decay.)
 - ▶ If **any** root of $A(s)$ has a positive real parts, then $Y(s)$ ultimately explodes. (Some term exponentially grows.)
 - ▶ What if $A(s)$ has a zero root (e.g., $G(s) = 1/s$)?
 - ▶ What if $A(s)$ has a purely imaginary root (e.g., $G(s) = 1/(s^2 + 1)$)?
- What if $A(s)$ has a purely imaginary root at $i\omega$, and $u(t) = \sin(\omega t)$ (for the exactly same ω)?

Effect of **poles** on **stability** and oscillations

- Definition of **stability**: We say that a system is “stable” if any **bounded** input signal gives a **bounded** output signal

“All poles are on LHP” is necessary for stability

The system $G(s)$ is stable if and only if **all** poles are on the left half plane (have negative real parts).

- Poles also affect whether the output response is oscillatory

“All poles are real” gives non-oscillatory step response

The step response of $G(s)$ is non-oscillatory if and only if all poles are real (have zero imaginary parts).

- Matlab: finding poles

```
p = poles(sys) % zeros(sys) for zeros
all(real(p) < 0) % if returns 1, then stable
all(imag(p) == 0) % if returns 1, then non-oscillatory
```


Static gain

- If the system is stable, then $K = G(0)$ is its static gain.

- Step response: When $u(t) = MS(t)$, $Y(s) = MG(s)/s$,

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} G(s) = G(0)M = KM.$$

- Matlab: finding static gain

$$K = \text{g.Numerator}\{1\}(\text{end}) / \text{g.Denominator}\{1\}(\text{end})$$

or simply $K = \text{dcgain}(\text{g})$

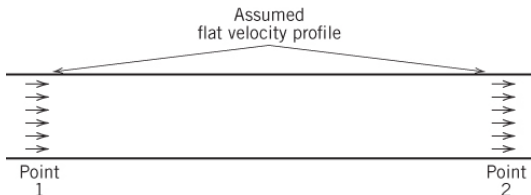
- ▶ “DC gain” = “direct current gain”

Finding poles, zeros, and gain [Seborg Ex.6.2]

$$G(s) = \frac{4(s+2)}{(0.2s+1)(2s+1)}$$

System with delay (deadtime)

- In industrial practice, people often use low-order models with a delay
 - ▶ First-order “plus” time delay (FOPTD) $G(s) = e^{-\theta s} \frac{K}{\tau s + 1}$
 - ▶ Second-order “plus” time delay (FOPTD) $G(s) = e^{-\theta s} \frac{K}{\tau^2 s^2 + 2\tau\zeta s + 1}$
- Physical picture: Plug flow tube



$$c_2(t) = c_1(t - \theta) \Rightarrow \frac{C_2(s)}{C_1(s)} = e^{-\theta s}$$

- ▶ Reason for time delay: **lack of mixing** makes the effect of input on the output **non-instantaneous**

** Approximating higher-order systems as FOPTD

Skogestad's half rule [Seborg Example 6.4]

$$G(s) = \frac{K(1-s)}{(12s+1)(3s+1)(0.2s+1)(0.05s+1)}e^{-s}$$

- 1 **Factorize** denominator. Keep the **slowest** term ($\tau = 12$).
- 2 Split the **second** slowest term into two halves ($3/2$). One half adds to $\tau \rightarrow 12 + 1.5$, the other half becomes a delay ($e^{-1.5s}$).
- 3 Approximate fast terms as delays ($e^{-0.2s}e^{-0.05s}$).
- 4 For the numerator terms, approximate as delays/anti-delays ($1 + \alpha s \approx e^{\alpha s}$, $1 - \alpha s \approx e^{-\alpha s}$) when $\alpha \ll \tau$. [Otherwise not approximable.]

$$G(s) \approx \frac{K}{13.5s+1}e^{-3.75s}.$$

** Approximating $e^{-\theta s}$ as a rational TF

- Padé approximation

$$e^{-\theta s} \approx R_{1,1}(s) = \frac{1 - \frac{\theta}{2}s}{1 + \frac{\theta}{2}s} \quad [1, 1]\text{-approximation}$$

$$e^{-\theta s} \approx R_{2,2}(s) = \frac{1 - \frac{\theta}{2}s + \frac{\theta^2}{12}s^2}{1 + \frac{\theta}{2}s + \frac{\theta^2}{12}s^2} \quad [2, 2]\text{-approximation}$$

- In general, Padé approximation $R_{m,n}(s)$ can be found by letting

$$\left. \frac{d^i R_{m,n}}{ds^i} \right|_{s=0} = \left. \frac{d^i G}{ds^i} \right|_{s=0}, \quad i = 0, 1, \dots, m+n$$

- ▶ For example, let $R_{1,1}(s) = \frac{b_0 + b_1 s}{1 + a_1 s}$ and then determine these 3 coefficients based on the above 3 equations.
- These approximations are usually not necessary, although they are often used just for simplicity. (Personal view: Simplification is desirable only if there is a good reason.)